Stat 122 (Mathematical Statistics II)

Lesson 3.1 Bias and Mean Square Error of an Estimator

Introduction

Probability distributions (discrete and continuous) are probability models. These models are used to describe populations of individuals. Below are examples.

- In a clinical trial with n patients, let p denote the probability of response to a new drug. A Bernoulli(p) model is assumed for each subject's response (e.g., respond/not).
- In an engineering application, the lifetime of an electrical circuit, Y, is under investigation. An exponential(β) model is assumed.
- In a public-health study, Y, the number of sexual partners in the past year, is recorded for a group of high-risk HIV patients. A Poisson(λ) model is assumed.
- In an ecological study, the amount of dead-weight (measured in g/plot), Y, is recorded. A $N(\mu, \sigma^2)$ model is assumed.

Each of these situations employs a probabilistic model that is indexed by population parameters. In real life, these parameters are unknown. An important statistical problem, thus, involves estimating these parameters with a random sample $Y_1, Y_2, \cdots, Y_n$, that is, an identically and identically distributed (*iid*) sample from the population. We can state this problem generally as follows.

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from a population which is described by the model $f_Y(y, \theta)$. Here, $f_Y(y, \theta)$ can either be a PMF or a PDF that describes the population of interest, and θ is a parameter that indexes the model. The statistical problem of interest is to estimate θ with the observed data $Y_1, Y_2, \cdots, Y_n$.

Definition

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from $f_Y(y, \theta)$. A **point estimator** $\hat{\theta}$ is a function of $Y_1, Y_2, \cdots, Y_n$ that estimates θ. Since $\hat{\theta}$ is (in general) a function of $Y_1, Y_2, \cdots, Y_n$, it is a *statistic*. In practice, θ could be a scalar or vector.

Example 3.1.1

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from a Poisson distribution with mean θ. We know that the probability mass function (PMF) for Y is given by

$$f_Y(y, \theta) = \begin{cases} \frac{\theta^y e^{-\theta}}{y!}, & y = 0, 1, 2, ... \\ 0, & \text{otherwise} \end{cases}$$

Here, the parameter is $\theta = E(Y)$. What estimator should we use to estimate θ?

Example 3.1.2

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid sample from a $U(0, \theta)$ distribution. We know that the probability density function (PDF) for Y is given by

$$f_Y(y, \theta) = \begin{cases} \frac{1}{\theta}, & 0 < y < \theta \\ 0, & \text{otherwise} \end{cases}$$

Here, the parameter is θ, the upper limit of the support of Y. What estimator should we use to estimate θ?

Example 3.1.3

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid sample from a $N(\mu, \sigma^2)$ distribution. We know that the probability density function (PDF) for Y is given by

$$f_Y(y, \theta) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}, & -\infty < y < \infty \\ 0, & \text{otherwise} \end{cases}$$

Here, the parameter is $\boldsymbol{\theta} = (\mu, \sigma^2)$, a vector of two parameters- the mean μ and the variance σ^2. What estimator should we use to estimate $\boldsymbol{\theta}$? Or, equivalently, we might ask how to estimate μ and σ^2 separately.

Bias and Mean Squared Error

In general, a *good* estimator $\hat{\theta}$ has the following properties:

1. $\hat{\theta}$ is unbiased for θ, and
2. $\hat{\theta}$ has small variance.

Definition

An estimator $\hat{\theta}$ is said to be unbiased for θ if

$$E(\hat{\theta}) = \theta$$

for all possible values of θ. If $E(\hat{\theta}) \neq \theta$ we say that $\hat{\theta}$ is a biased estimator for θ.

In general, the bias of an estimator is

$$B(\hat{\theta}) \equiv E(\hat{\theta}) - \theta$$

If $B(\hat{\theta}) > 0$, then $\hat{\theta}$ overestimates θ. If $B(\hat{\theta}) < 0$, then $\hat{\theta}$ underestimates θ. If $\hat{\theta}$ is unbiased, then, of course, $B(\hat{\theta}) = 0$.

Example 3.1.1: Revisited!

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample from a Poisson distribution with mean θ. Recall that, in general, the sample mean $\overline{Y}$ is an unbiased estimator for a population mean μ. For the Poisson model, the (population) mean is $\mu = E(Y) = \theta$. Thus, we know that

$$\hat{\theta} = \overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

is an unbiased estimator of θ.

Recall also that the variance of the sample mean, $V(\overline{Y})$, is, in general, the population variance σ^2 divided by n. For the Poisson model, the (population) variance is $\sigma^2 = \theta$; thus, $V(\hat{\theta}) = V(\overline{Y}) = \theta/n$.

Example 3.1.2: Revisited!

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid sample from a $U(0, \theta)$ distribution. Consider the point estimator $Y_{(n)}$. Intuitively, this seems like a reasonable estimator to use, the largest order statistic should be fairly close to θ), the upper endpoint of the support of Y.

To compute $E(Y_{(n)})$, we have to know the PDF of $Y_{(n)}$. We learned from Module 1 that

$$\begin{aligned} f_{y_{(n)}}(y) &= n f_Y(y)[F_Y(y)]^{n-1} \\ &= n\left(\frac{1}{\theta}\right)\left[\frac{y}{\theta}\right]^{n-1} \\ &= n\theta^{-n}y^{n-1} \end{aligned}$$

Thus,

$$
\begin{aligned}
E(Y_{(n)}) &= \int_0^\theta y \times n\theta^{-n} y^{n-1}\, dy \\
&\vdots \\
&= \left(\frac{n}{n+1}\right)\theta
\end{aligned}
$$

We see that $Y_{(n)}$ is a biased estimator of θ (it underestimates θ on average).

Suppose we modify (transform) $Y_{(n)}$ to come up with a new estimator that is unbiased for θ. In fact, it can be shown that

$$\hat{\theta} = \left(\frac{n+1}{n}\right) Y_{(n)}$$

is unbiased for θ.

CHALLENGE QUESTION: What is the variance of $\hat{\theta}$?

<u>Example 3.1.3</u>: Revisited!

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid sample from a $N(\mu, \sigma^2)$ distribution. To estimate μ, we know that a good estimator is $\overline{Y}$. The sample mean $\overline{Y}$ is unbiased, that is, $E(\overline{Y}) = \mu$ and, furthermore, $V(\overline{Y}) = \sigma^2/n$ decreases as the sample size n increases.

To estimate σ^2, we can use the sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$$

For the normal model, the sample variance is unbiased. To see this, recall that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

so that

$$E\left[\frac{(n-1)S^2}{\sigma^2}\right] = n-1$$

since the mean of a χ^2 random variable equals its degrees of freedom. Thus,

$$
\begin{aligned}
&\implies n-1 = E\left[\frac{(n-1)S^2}{\sigma^2}\right] \\
&\implies n-1 = \left(\frac{n-1}{\sigma^2}\right) E(S^2) \\
&\implies E(S^2) = \sigma^2
\end{aligned}
$$

showing that S^2 is an unbiased estimator of the population variance σ^2.

CHALLENGE QUESTION: What is the variance of S^2?

In some problems, the goal is to estimate a function of θ, say, $g(\theta)$. The following example illustrates how we can find unbiased estimator of a function of θ.

Example 3.1.4

Suppose that $Y_1, Y_2, \cdots, Y_n$ are iid exponential observations with mean θ. Derive an unbiased estimator for $g(\theta) = \frac{1}{\theta}$.

SOLUTION:

Since $E(Y) = \theta$, one's intuition might suggest to try $\frac{1}{\overline{Y}}$ as an estimator for $\frac{1}{\theta}$. First, note that

$$E\left(\frac{1}{\overline{Y}}\right) = nE\left(\frac{1}{T}\right)$$

where $T = \sum_{i=1}^{n} Y_i$. Recall that $Y_1, Y_2, \cdots, Y_n$ iid exponential(θ) then $T \sim Gamma(n; \theta)$, therefore

$$\begin{aligned} E\left(\frac{1}{\overline{Y}}\right) &= nE\left(\frac{1}{T}\right) \\ &= n\int_0^\infty \frac{1}{t}\frac{1}{\Gamma(n)\theta^n} t^{n-1} e^{t/\theta}\, dt \\ &\vdots \\ &= \left(\frac{n}{n-1}\right)\frac{1}{\theta} \end{aligned}$$

This shows that $1/\overline{Y}$ is a biased estimator of $1/\theta$. However,

$$\begin{aligned} E\left[\left(\frac{n-1}{n}\right)\frac{1}{\overline{Y}}\right] &= \left(\frac{n-1}{n}\right) E\left[\frac{1}{\overline{Y}}\right] \\ &= \left(\frac{n-1}{n}\right)\left(\frac{n}{n-1}\right)\frac{1}{\theta} \\ &= \frac{1}{\theta} \end{aligned}$$

This shows that

$$\left(\frac{n-1}{n}\right)\frac{1}{\overline{Y}}$$

is unbiased for $1/\theta$.

Definition

The Mean Squared Error (MSE) of a point estimator $\hat{\theta}$ is given by

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = V(\hat{\theta}) + [B(\hat{\theta})]^2$$

We see that the MSE combines the

- the precision (variance) of $\hat{\theta}$, and
- accuracy (bias) of $\hat{\theta}$.

REMARKS:

1. Of course, if $\hat{\theta}$ is unbiased for θ, then $MSE(\hat{\theta}) = V(\hat{\theta})$, since $B(\hat{\theta}) = 0$.
2. Suppose that we have two unbiased estimators, say, $\hat{\theta}_1$ and $\hat{\theta}_2$. Then we would prefer to use the one with the smaller variance. That is, if $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, then we would prefer $\hat{\theta}_1$ as an estimator. Note that it only makes sense to choose an estimator on the basis of its variance when both estimators are unbiased.
3. Suppose that we have two biased estimators, say, $\hat{\theta}_1$ and $\hat{\theta}_2$. We choose the estimator with the smaller MSE. That is, if $MSE(\hat{\theta}_1) < MSE(\hat{\theta}_2)$, then we would prefer $\hat{\theta}_1$ as an estimator.

<u>Example 3.1.5</u>

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* Bernoulli(p) sample, where $0 < p < 1$. Define $X = \sum_{i=1}^{n} Y_i$. Consider the following two estimators

$$\hat{p}_1 = \frac{X}{n}$$

and

$$\hat{p}_2 = \frac{X+2}{n+4}$$

Which estimator should we use to estimate p?

SOLUTION

First, note that $X \sim B(n, p)$. Thus,

$$\begin{aligned} E(\hat{p}_1) &= E\left[\frac{X}{n}\right] \\ &= \frac{1}{n}E(X) \\ &= \frac{1}{n}(np) \\ &= p \end{aligned}$$

Thus, $\hat{p}_1$ is unbiased for p.

Meanwhile,

$$\begin{aligned} E(\hat{p}_2) &= E\left[\frac{X+2}{n+4}\right] \\ &= \frac{1}{n+4}E[X+2] \\ &= \frac{1}{n+4}(np+2) \end{aligned}$$

This shows that $\hat{p}_2$ is biased for p.

Thus, to compare $\hat{p}_1$ and $\hat{p}_2$ as estimators, we should use the estimators' MSE since $\hat{p}_2$ is biased.

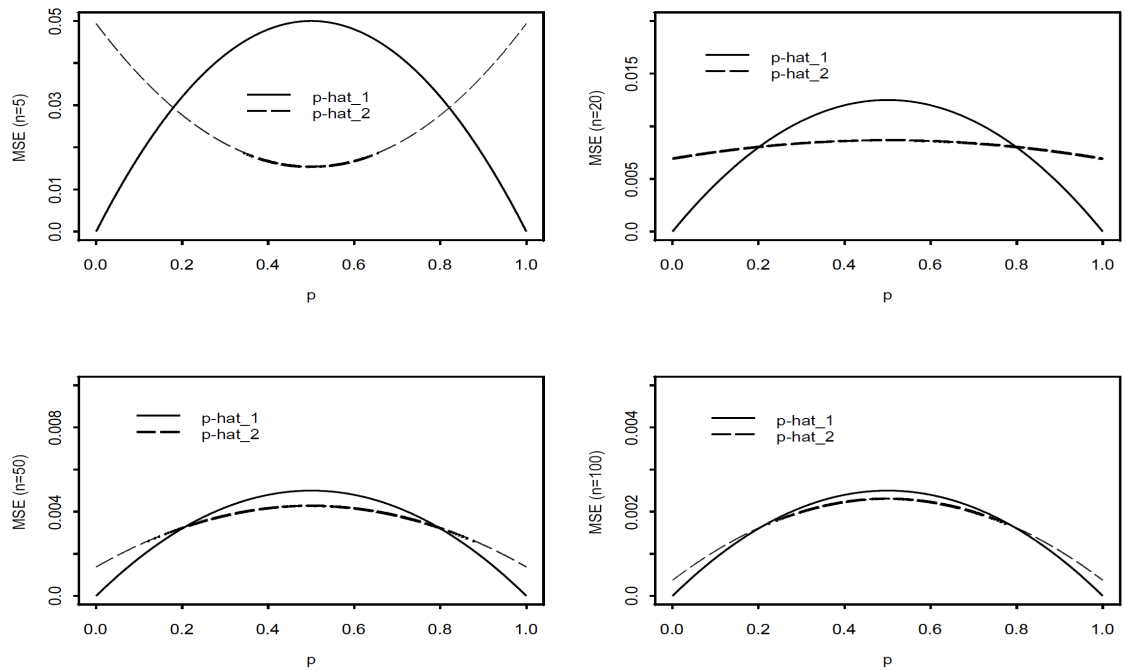
Since $\hat{p}_1$ is unbiased for p, then its MSE is just equal to its variance. That is,

$$\begin{aligned} MSE(\hat{p}_1) &= V(\hat{p}_1) \\ &= V\left(\frac{X}{n}\right) \\ &= \frac{1}{n^2}V(X) \\ &= \frac{1}{n^2}[np(1-p)] \\ &= \frac{p(1-p)}{n} \end{aligned}$$

Meanwhile, the MSE of $\hat{p}_2$ is

$$\begin{aligned} MSE(\hat{p}_2) &= V(\hat{p}_2) + [B(\hat{p}_2)]^2 \\ &\vdots \\ &= \frac{np(1-p)}{(n+4)^2} + \left(\frac{np+2}{n+4} - p\right)^2 \end{aligned}$$

Below are plots of the MSE of $\hat{p}_1$ and $\hat{p}_2$ for various sample sizes ($n = 5, 10, 20, 100$)



We can see that neither estimator is uniformly superior, that is, neither estimator delivers a smaller MSE for all $0 < p < 1$. However, for smaller sample sizes, $\hat{p}_2$ often beats $\hat{p}_1$ (in terms of MSE) when p is in the vicinity of 0.5; otherwise, $\hat{p}_1$ often provides smaller MSE.